Digitaler Zwilling für eine (virtuelle) Produktionsanlage

Programmierpraktikum 2025

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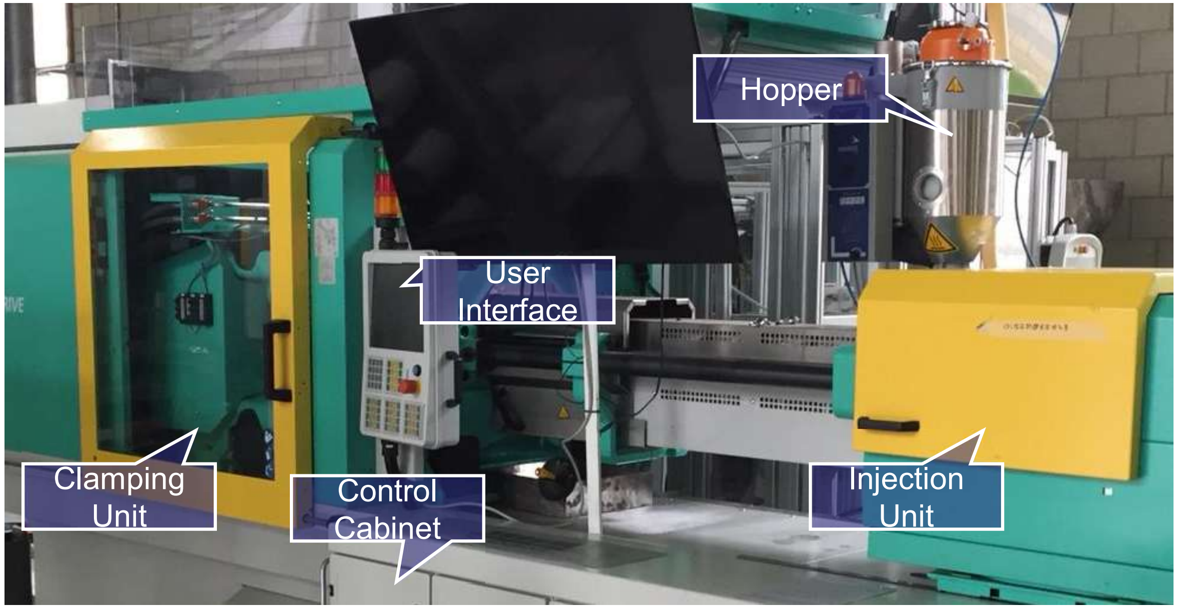
Sommersemester 2025

Project Scope

- Complex production systems

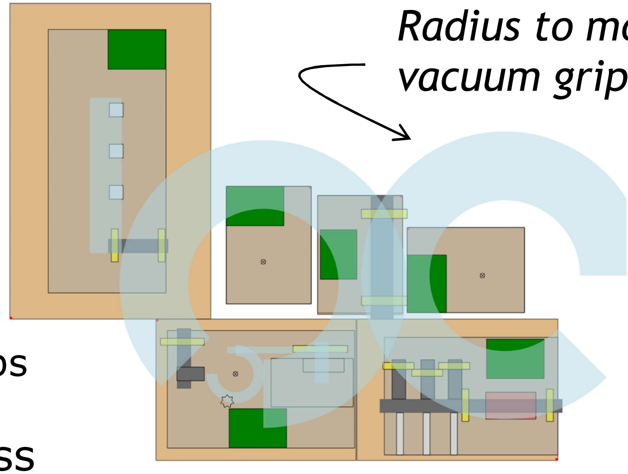
- Challenges
  - Highly relevant *context*
  - *Large amount* of *heterogeneous* data
  - *Long life span*
  - *Real-time* updates
  - *Changing* reality

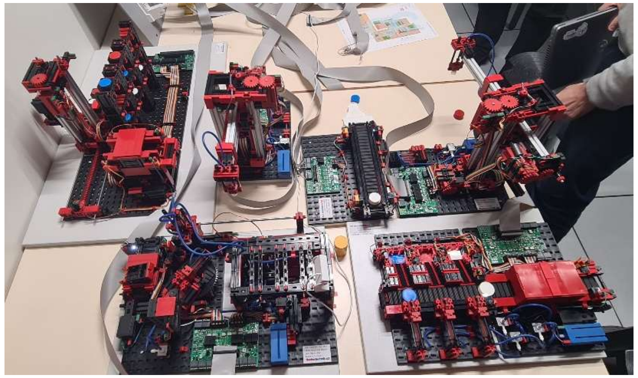
- Create *digital twins* (=software systems) to
  - monitor their behavior
  - analyze processes
  - optimize steps
  - predict failures
  - …



A production process „in the small" | Fischertechnik

- Production of, e.g., metal blanks
  - Production line with processing stages
  - Blanks in various shapes
    - with threaded hole
    - with round bores
  - Deformed
  - Come in different colors
  - Can be heated for further processing steps

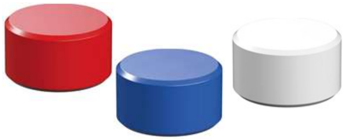




- Different views and roles on the process
  - Users can view goods, place orders, view the status of orders
  - Monitoring for purchasing: material stock, throughput, demand prediction
  - Monitoring of the factory: capacity utilization, status, errors

For now only virtually at Regensburg…
... already existing in Aachen, Stuttgart, Rennes

Fischertechnik | Workpieces, transport, handling

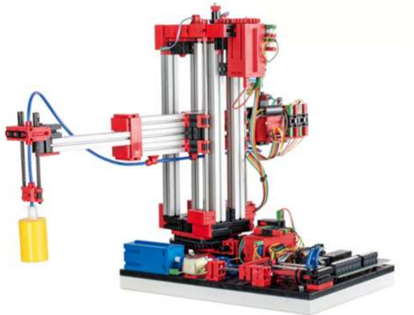


workpieces

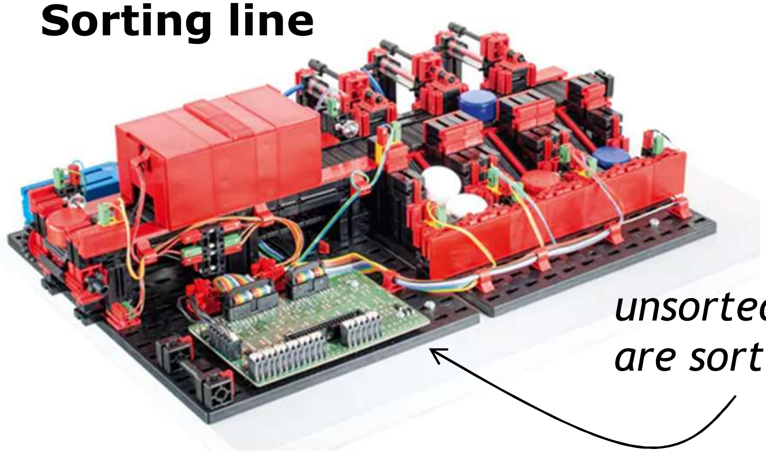
Conveyor belt

3D gripper

Vacuum gripper

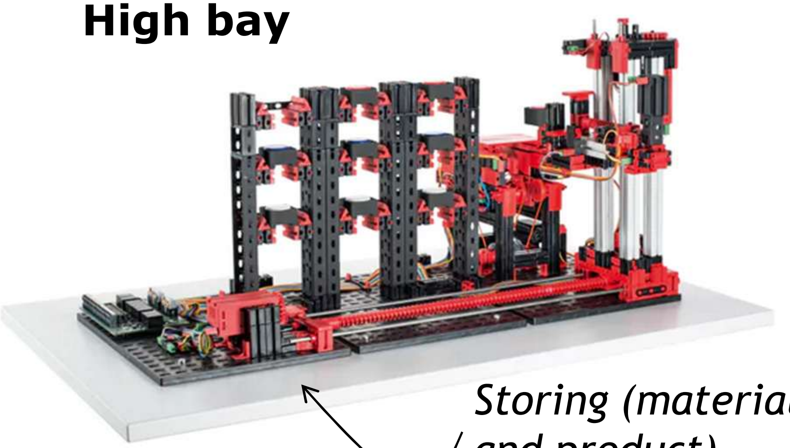


Sorting line



unsorted workpieces are sorted and counted

High bay



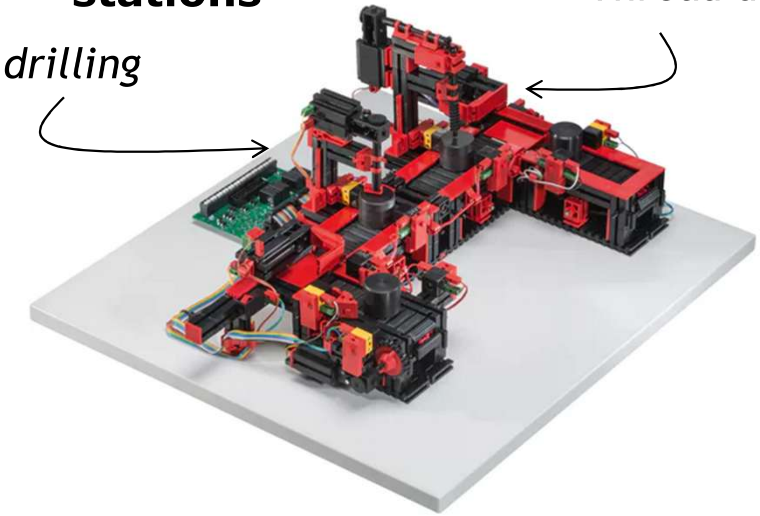
Storing (material and product)

Fischertechnik | Machines

Indexed line with 2 machining stations

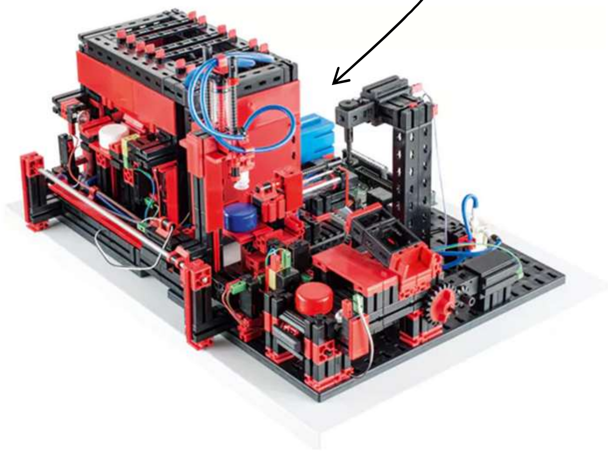
Round drilling

Thread drilling

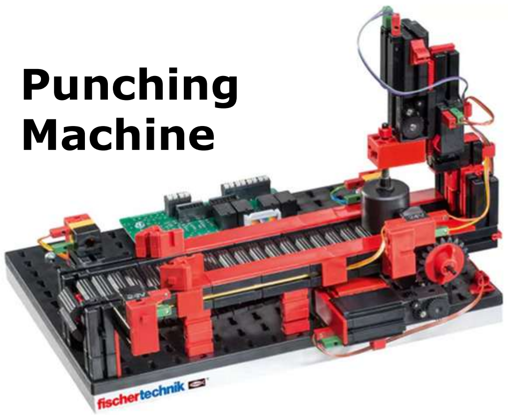


Multiprocessing station

Heating



Punching Machine



Deformation

Background

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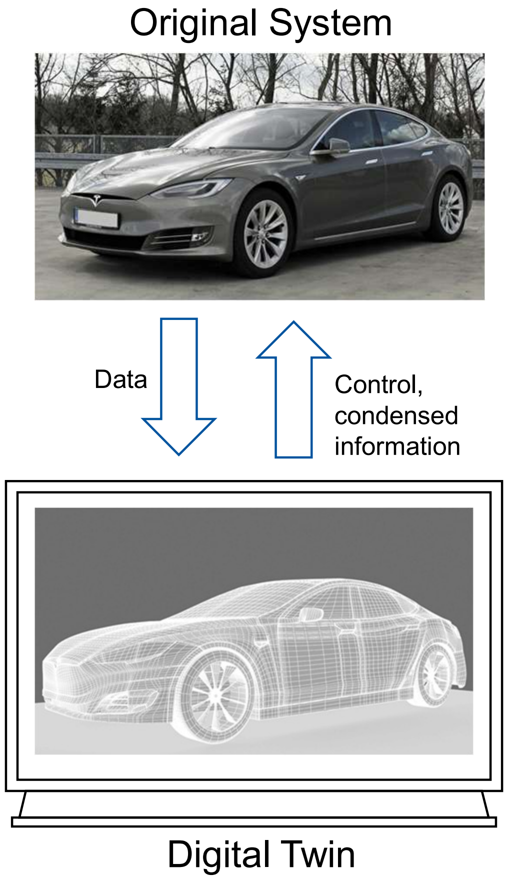
Digital Twins as *complex, long-lasting, software systems*



contextual data and their aggregation and abstraction

A Digital Twin of an original system is a software that consists of

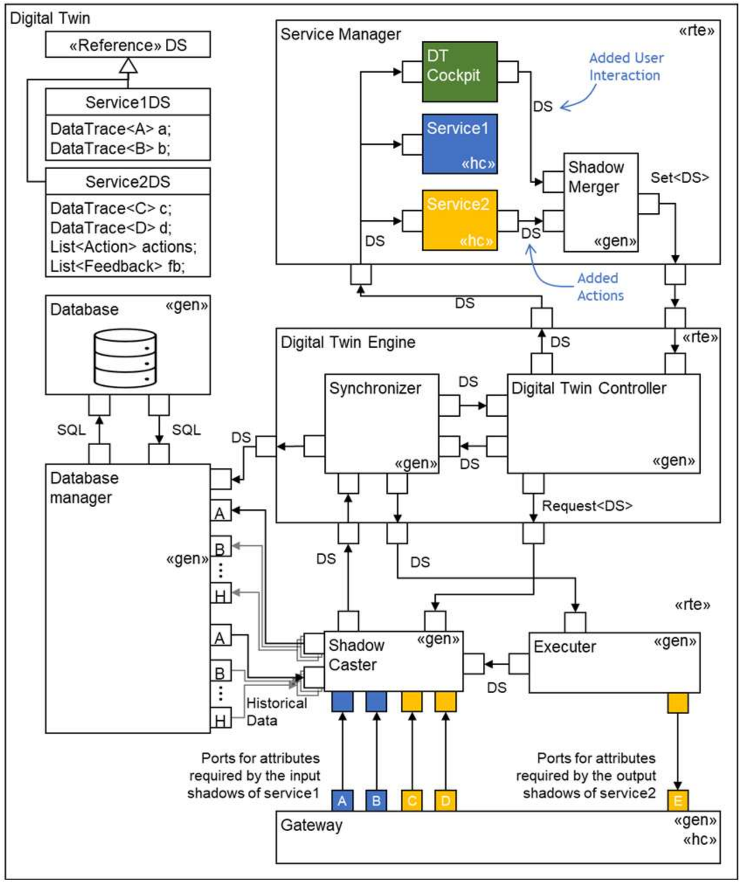
- a set of models of the system and
- a set of digital shadows,
  - both of which are purposefully updated on a regular basis, and
- provides a set of services to use both purposefully with respect to the original system.

- The digital twin interacts with the original system by
  - providing useful information about the system's context and
  - sending it control commands.

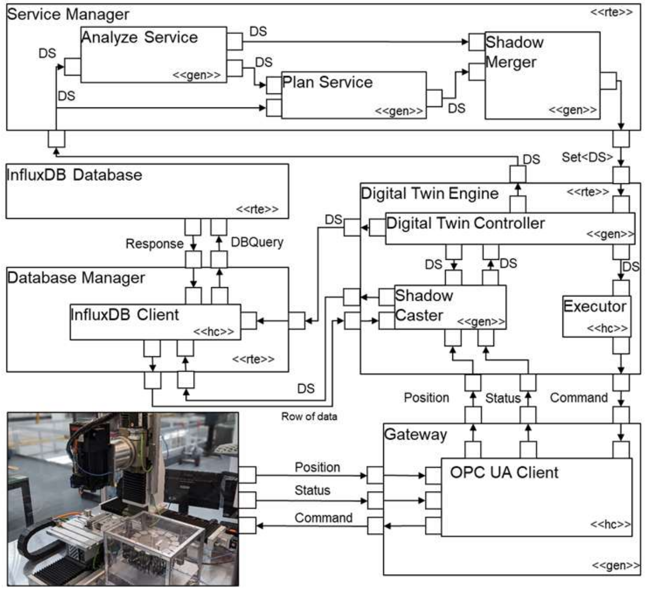
B. Rumpe, J. Michael, A. Wortmann: Digital Twins 2.2. https://www.se-rwth.de/essay/Digital-Twin-Definition/

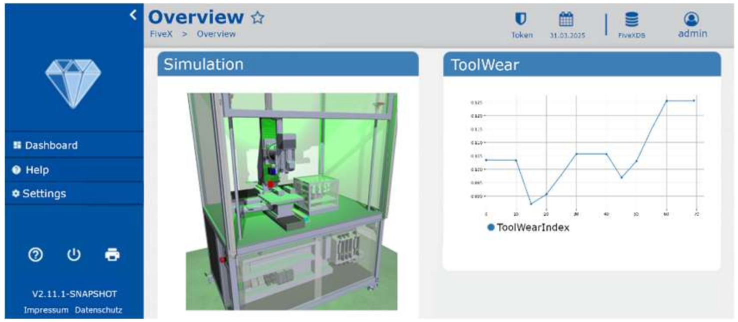
Architecture

Reference Architecture



Concrete Implementation for a Milling machine



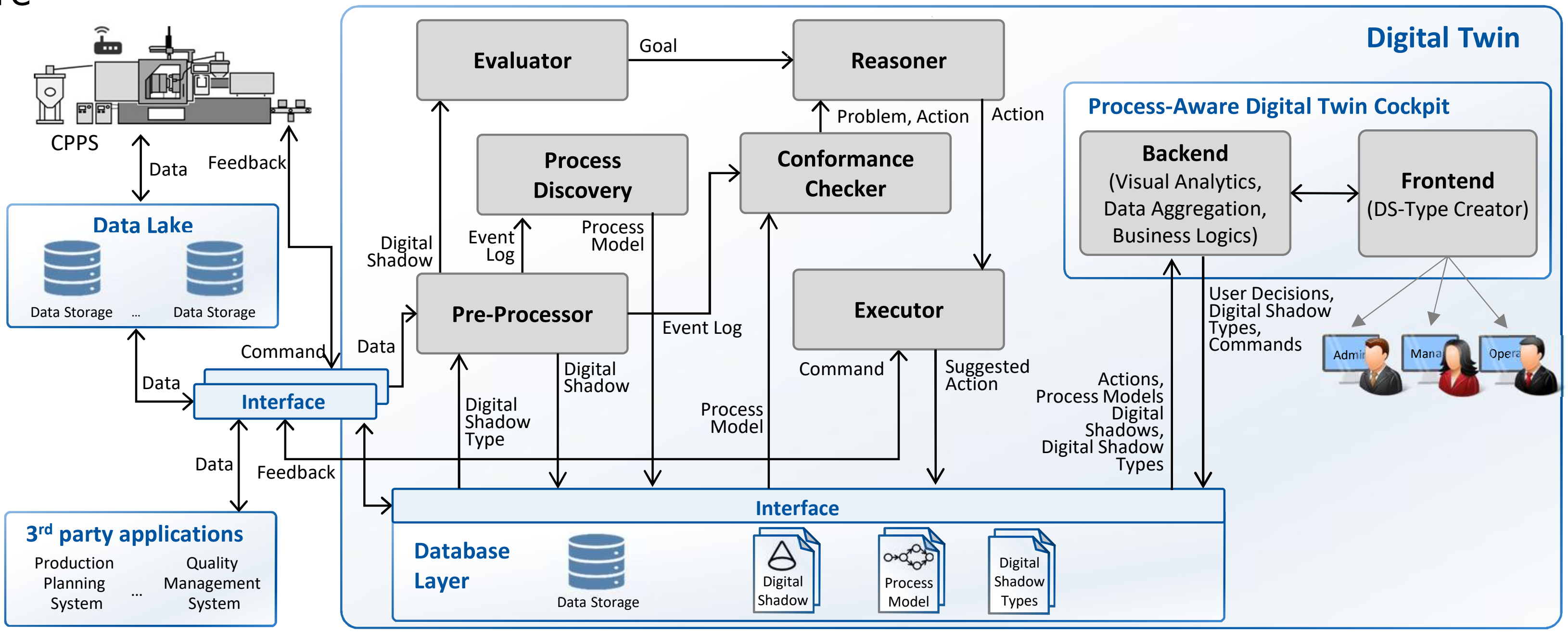


Example Architecture | Process Prediction with Digital Twins [BHK+21]

Self-adaptive DT architecture
- MAPE-K Loop

Aim:
improve the DT operation with additional services
- process discovery from event logs and
- process prediction from process models at runtime

- Models at *design and runtime*
  - domain model, GUI models, architecture, digital shadow structure [BBD+21]
  - process models, goals, actions,…



[BHK+21] T. Brockhoff, M. Heithoff, I. Koren, J. Michael, J. Pfeiffer, B. Rumpe, M.S. Uysal, W.M.P. van der Aalst, A. Wortmann: *Process Prediction with Digital Twins*. Models@run.time WS (MODELS'21)
[BBD+21] F. Becker, P. Bibow, M. Dalibor, A. Gannouni, V. Hahn, C. Hopmann, M. Jarke, I. Koren, M. Kröger, J. Lipp, J. Maibaum, J. Michael, B. Rumpe, P. Sapel, N. Schäfer, G. J. Schmitz, G. Schuh, and A. Wortmann: *A conceptual model for digital shadows in industry and its application*. ER'21

Literature (as starting points and for further information)

- [MCZ+24] J. Michael, L. Cleophas, S. Zschaler, T. Clark, B. Combemale, T. Godfrey, D.E. Khelladi, V. Kulkarni, D. Lehner, B. Rumpe, M. Wimmer, A. Wortmann, S. Ali, B. Barn, I. Barosan, N. Bencomo, F. Bordeleau, G. Grossmann, G. Karsai, O. Kopp, B. Mitschang, P. Muñoz Ariza, A. Pierantonio, F. A. C. Polack, M. Riebisch, H. Schlingloff, M. Stumptner, A. Vallecillo, M. van den Brand, H. Vangheluwe: *Model-Driven Engineering for Digital Twins: Opportunities and Challenges.* INCOSE Systems Engineering, 2025.
  - Overview: What is a DT, what to use models for
- M. Heithoff, J. Michael, B. Rumpe, J. Pfeiffer, A. Wortmann, J. Zhang: *A Method for Model-Driven Engineering of Digital Twins in Manufacturing*. (In review)
  - Architecture and examples from the architecture slide
- [DMR+20] M. Dalibor, J. Michael, B. Rumpe, S. Varga, A. Wortmann: *Towards a Model-Driven Architecture for Interactive Digital Twin Cockpits.* In: Conceptual Modeling (ER'20), Springer, 2020.
  - A DT architecture, component descriptions
- [HJM+24] M. Heithoff, N. Jansen, J. Michael, F. Rademacher, B. Rumpe: *Model-Based Engineering of Multi-Purpose Digital Twins in Manufacturing*. In: Digital Twin - Fundamentals and Applications. Springer, 2024.
  - Example use cases and services from injection molding
- [CJJ+23] B. Combemale, N. Jansen, J.- M. Jézéquel, J. Michael, Q. Perez, F. Rademacher, B. Rumpe, D. Vojtisek, A. Wortmann, J. Zhang: *Model-Based DevOps: Foundations and Challenges.* MoDDiT Workshop 2023.
  - DevOps for DTs and sustainability related questions
- [KMR+20] J.C. Kirchhof, J. Michael, B. Rumpe, S. Varga, and A. Wortmann: *Model-driven digital twin construction: synthesizing the integration of cyber-physical systems with their information systems.* MODELS '20. https://doi.org/10.1145/3365438.3410941
  - Connecting DT and IoT systems in a model-based implementation
- [DJR+22] M. Dalibor, N. Jansen, B. Rumpe, D. Schmalzing, L. Wachtmeister, M. Wimmer, A. Wortmann: *A cross-domain systematic mapping study on software engineering for Digital Twins.* In: Journal of Systems and Software (JSS) 193, Elsevier, 2022. https://doi.org/10.1016/j.jss.2022.111361
  - Overview of DT usages and services

Tasks & Process

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What you should realize

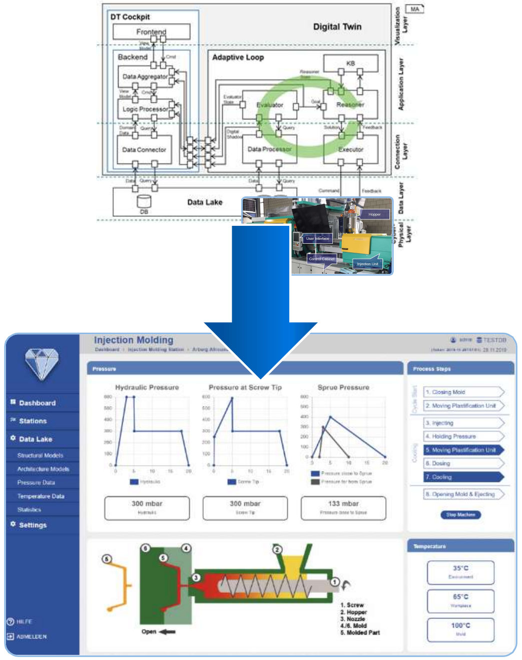
- Develop a digital twin
  - Client-server application

  - *Representation* of the production line
    - Models, e.g., describing the physical setup (SysML)
    - Data (real and simulated)

  - *Services*
    - Basic components (handle data, DT engine for orchestration,…)
    - Advanced services for analysis (based on selected use case)

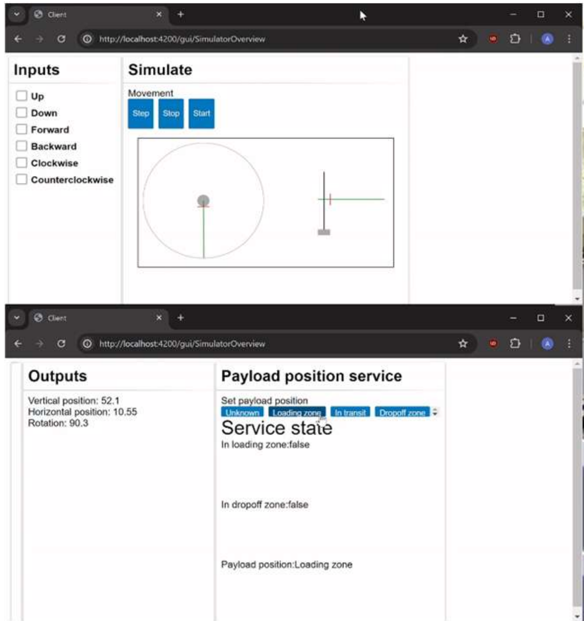
  - *Dashboard*
    - Visualization of the data, models and processes

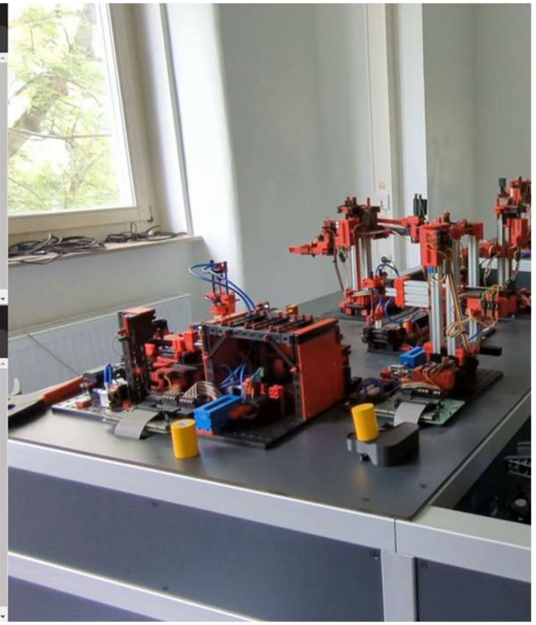
  - *Connection* to the physical system
    - Mocks for sensors and actuators



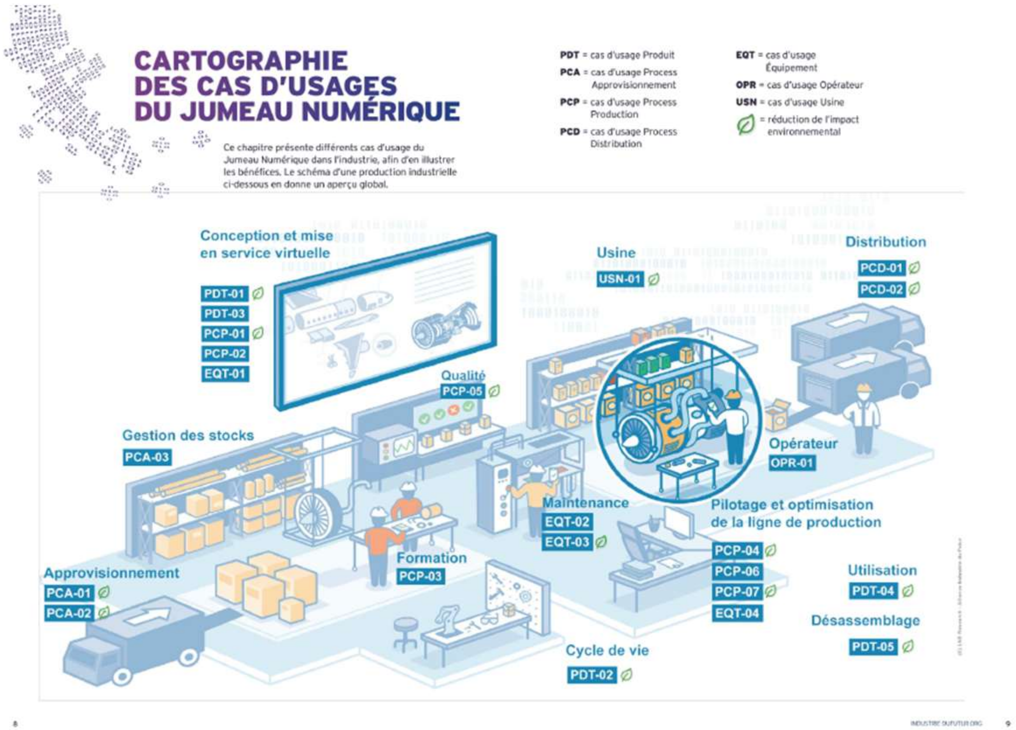
What you will get

- Use Cases
  - Choose some interesting services
- Description of the machine interfaces
  - MQTT
  - Machine input/output
- Example data from a real setup
  - Machine input/output (MQTT data)
  - Record entry play
- 2D models of machines





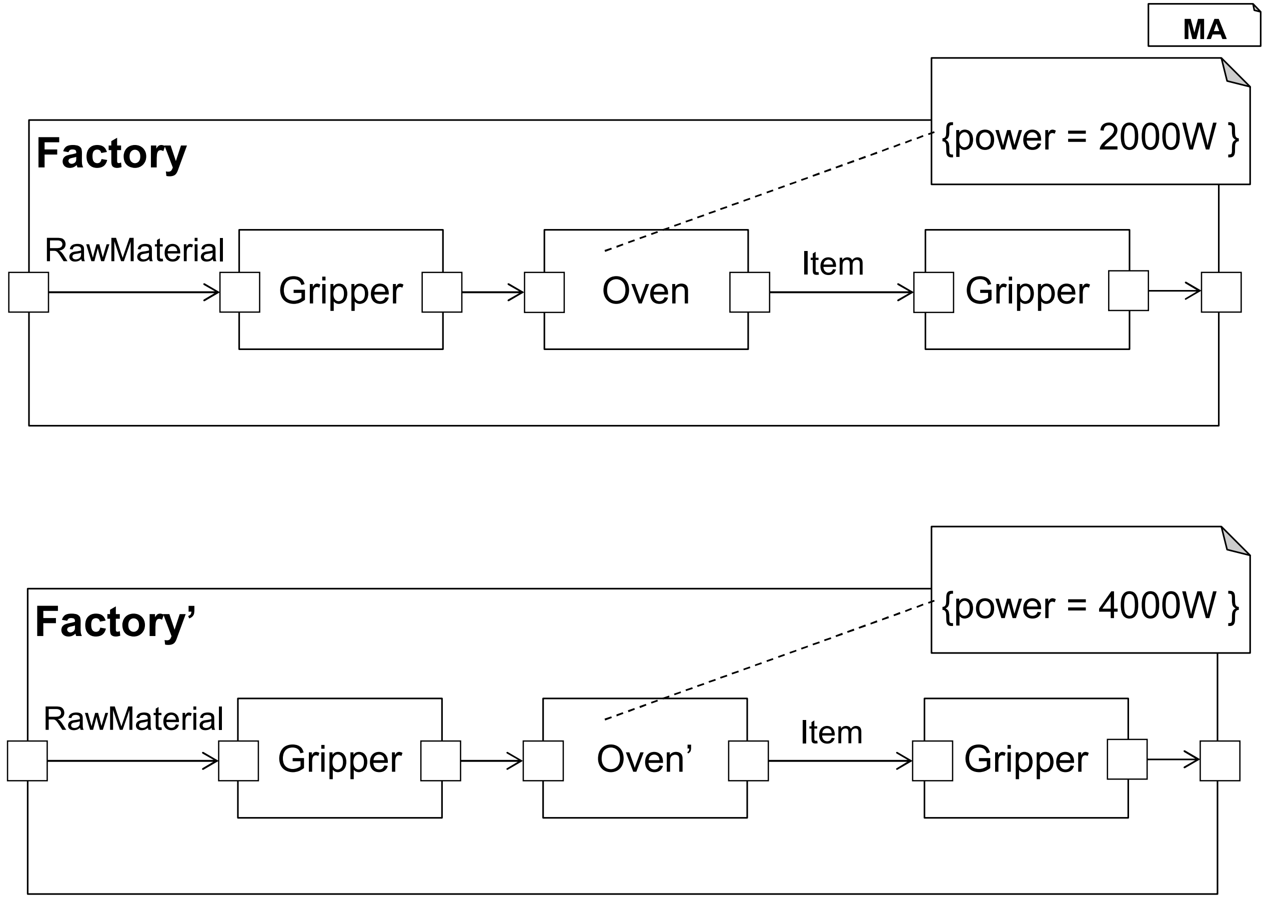
Use Cases



- Monitoring the status of the production line (mandatory)
- Analysis
  - E.g., energy usage, errors, throughput,…
- Optimization
  - E.g., energy usage, speed, orders, …
- Simulating various scenarios:
  - E.g., loss of a redundant machine and recalibration of the process, …
- Prediction
  - E.g., when will a part break, should be replaced,…

Example DT Service | Scenario Based Simulation

- Optimize efficiency of a system
  - measuring resources and output
  - optimization
- Service: Scenario Based Simulation
  - Use models with estimated resource consumption + output to estimate efficiency
  - By comparing different scenarios the system can be optimized w.r.t.
    - Efficiency
    - Fault tolerance
    - …
- Feed measured data back into the DT to improve the estimations



Fischertechnik – Einige beschreibende Modelle

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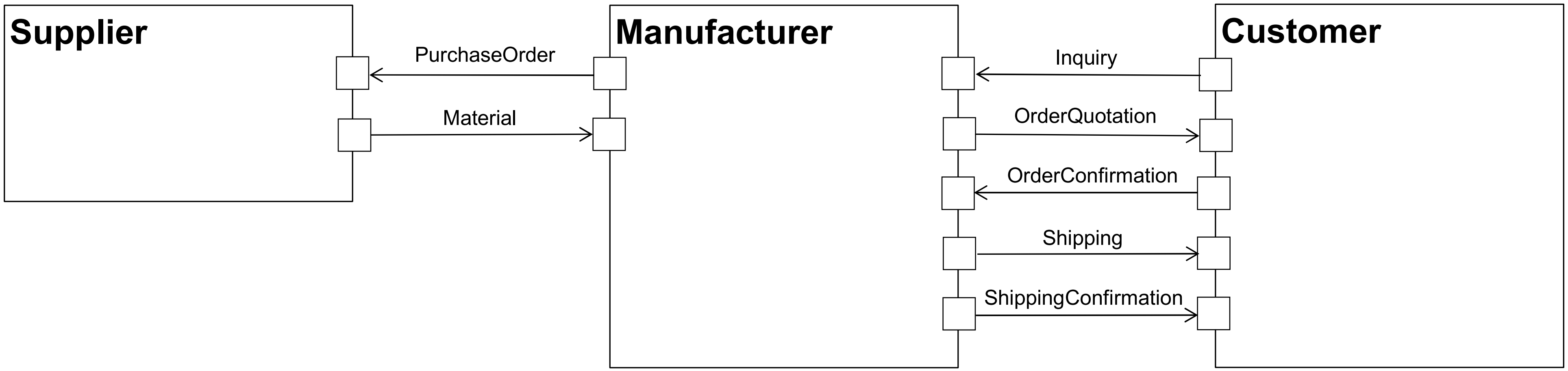
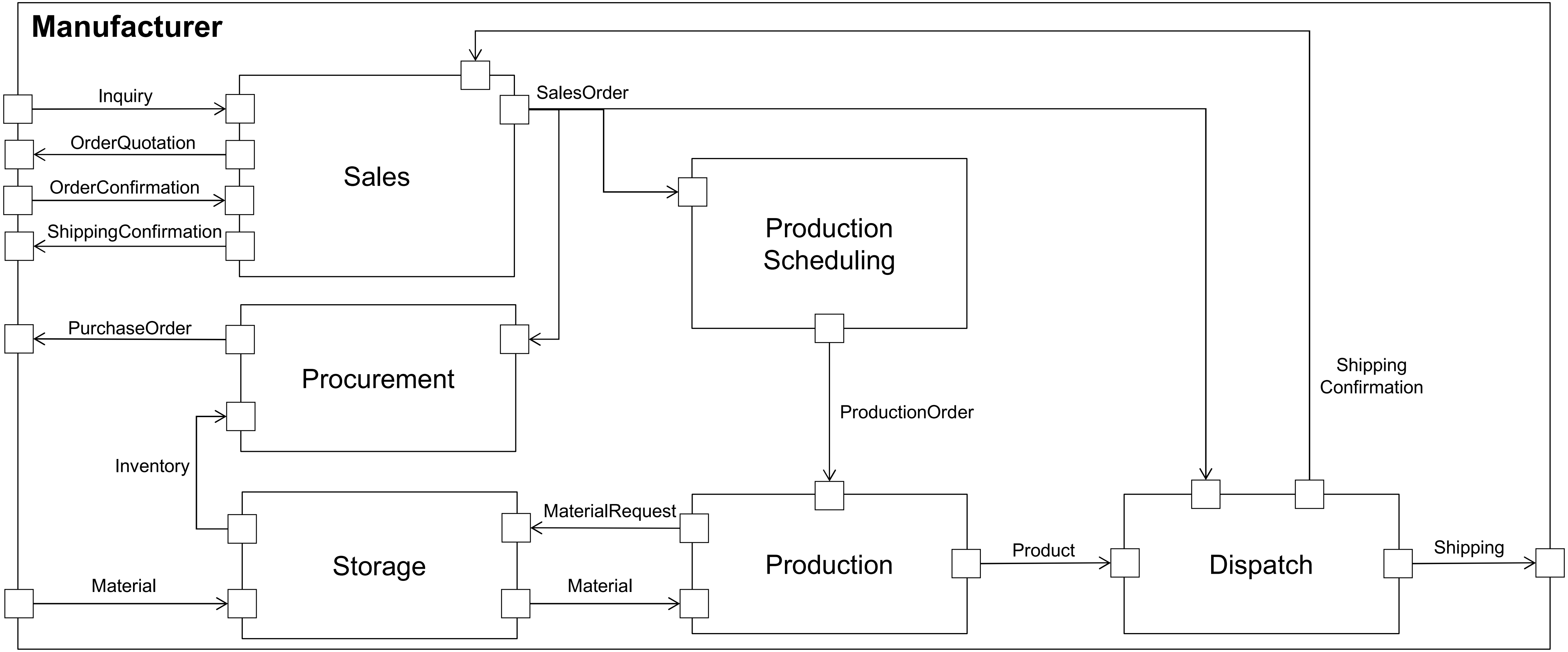
System

MA


Supplier
Manufacturer
Customer
PurchaseOrder
Material
Inquiry
OrderQuotation
OrderConfirmation
Shipping
ShippingConfirmation

Manufacturer

MA



Manufacturing Types

- Make-to-stock (MTS): Production of the parts is decoupled from the direct demand of customers.
- **Make-to-Order (MTO): The part‘s manufacturing only starts if the customer places an order.**
- Assemble-to-order (ATO): Combines MTS and MTO.
- Engineer-to-order (ETO): Part that the manufacturer does not have in the portfolio.

Patrick Sapel, Christian Hopmann, Towards an ontology-based dictionary for production planning and control in the domain of injection molding as a basis for standardized asset administration shells
https://doi.org/10.1016/j.jii.2023.100488

Production

1. The sales order contains the part(s) to manufacture, the costs, and the due date, but also information about the priority and status of the sales order.
2. Within the scheduling step, the sales order is transferred to one or more production orders.
3. A production order initiates the production of a part and consists of a bill of materials and production steps.
   - The BOM lists the material and its amount for production.
   - Production steps are the single operations necessary for finishing the product.
4. As soon as all material is available, the production starts.
5. After the production is finished the order is fulfilled.

Patrick Sapel, Christian Hopmann, Towards an ontology-based dictionary for production planning and control in the domain of injection molding as a basis for standardized asset administration shells
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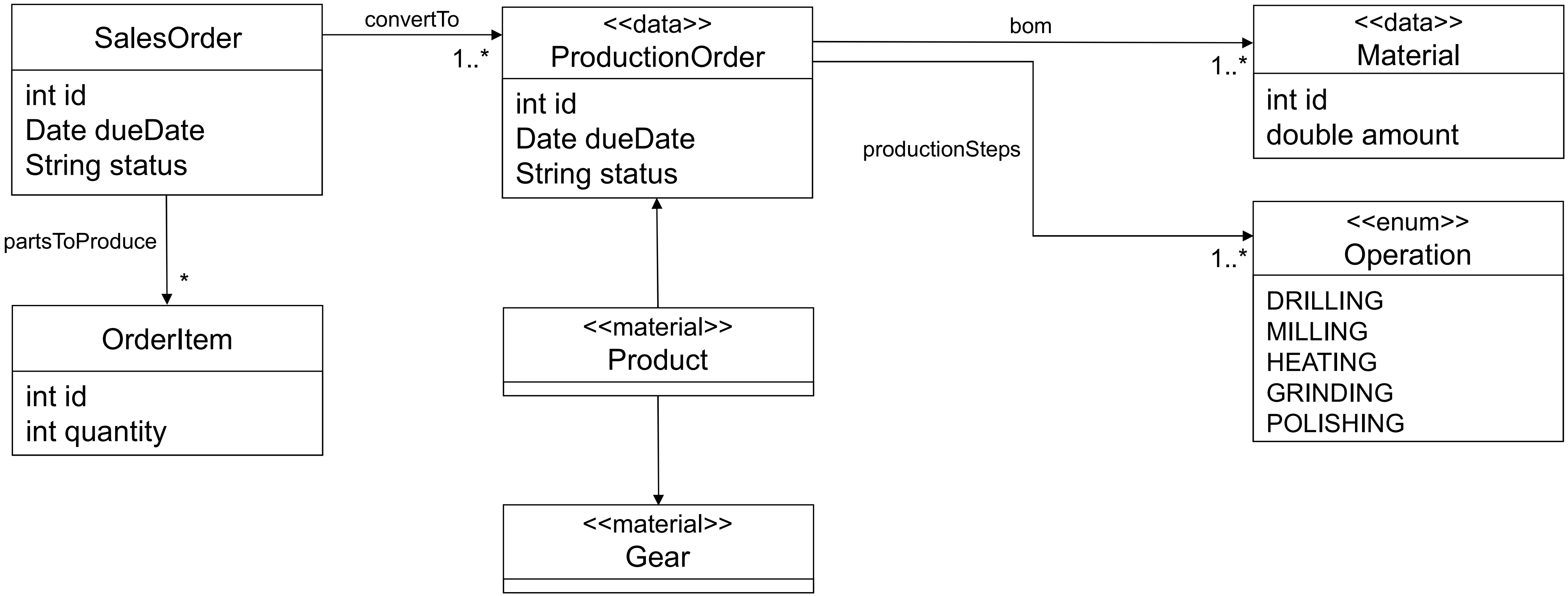
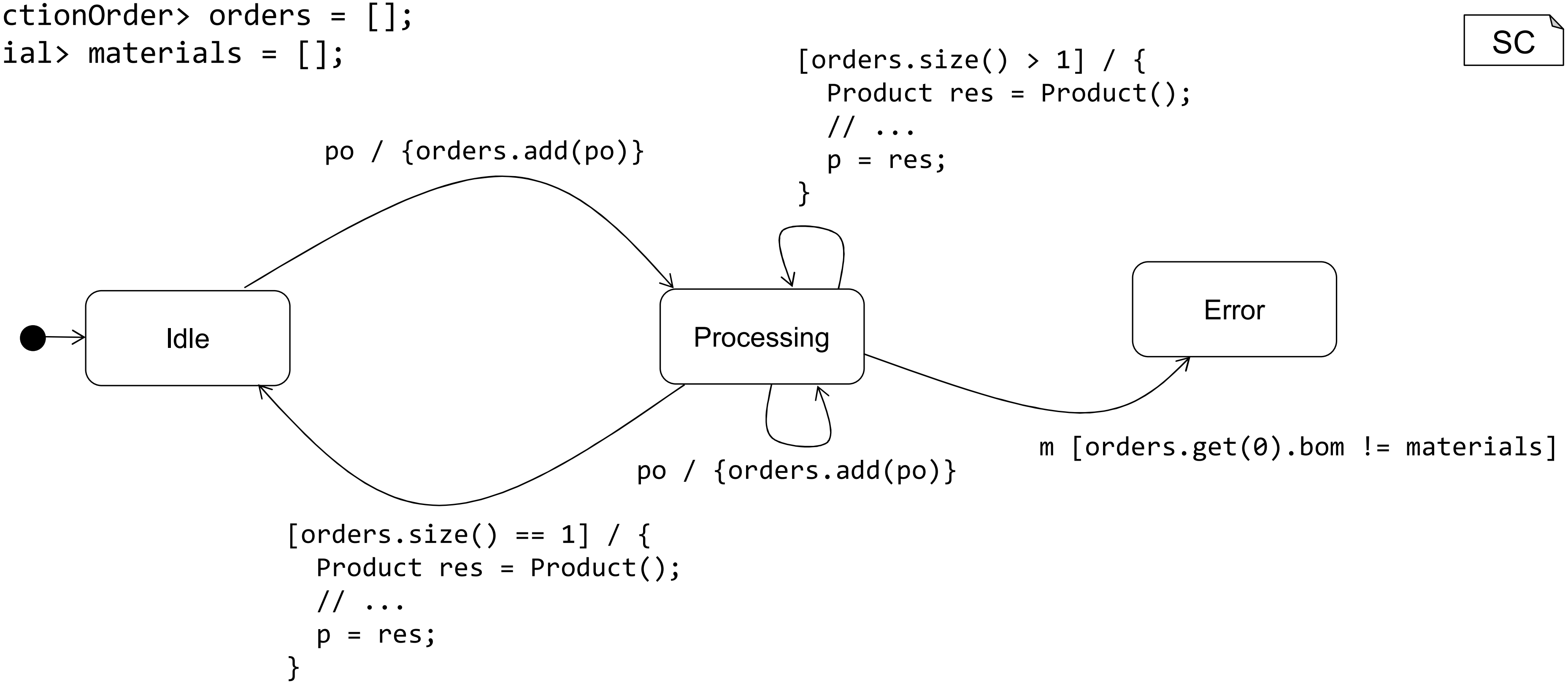
Production

CD



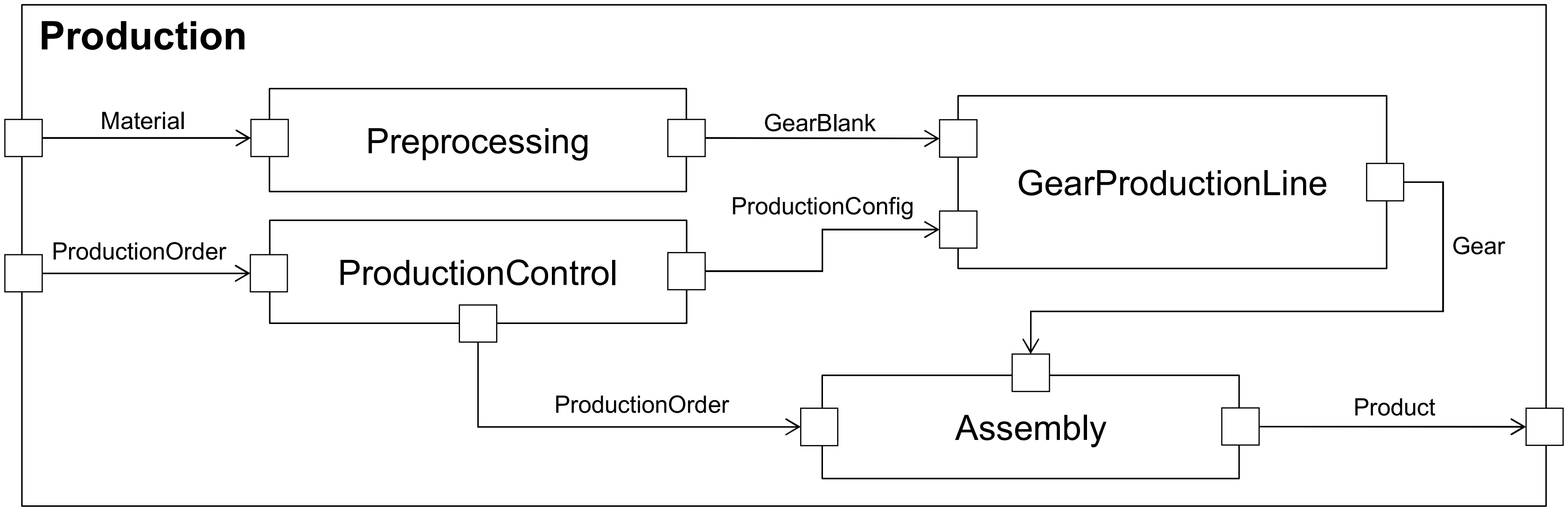
Production

Input: ProductionOrder po, Material m
Output: Product p
List<ProductionOrder> orders = [];
List<Material> materials = [];

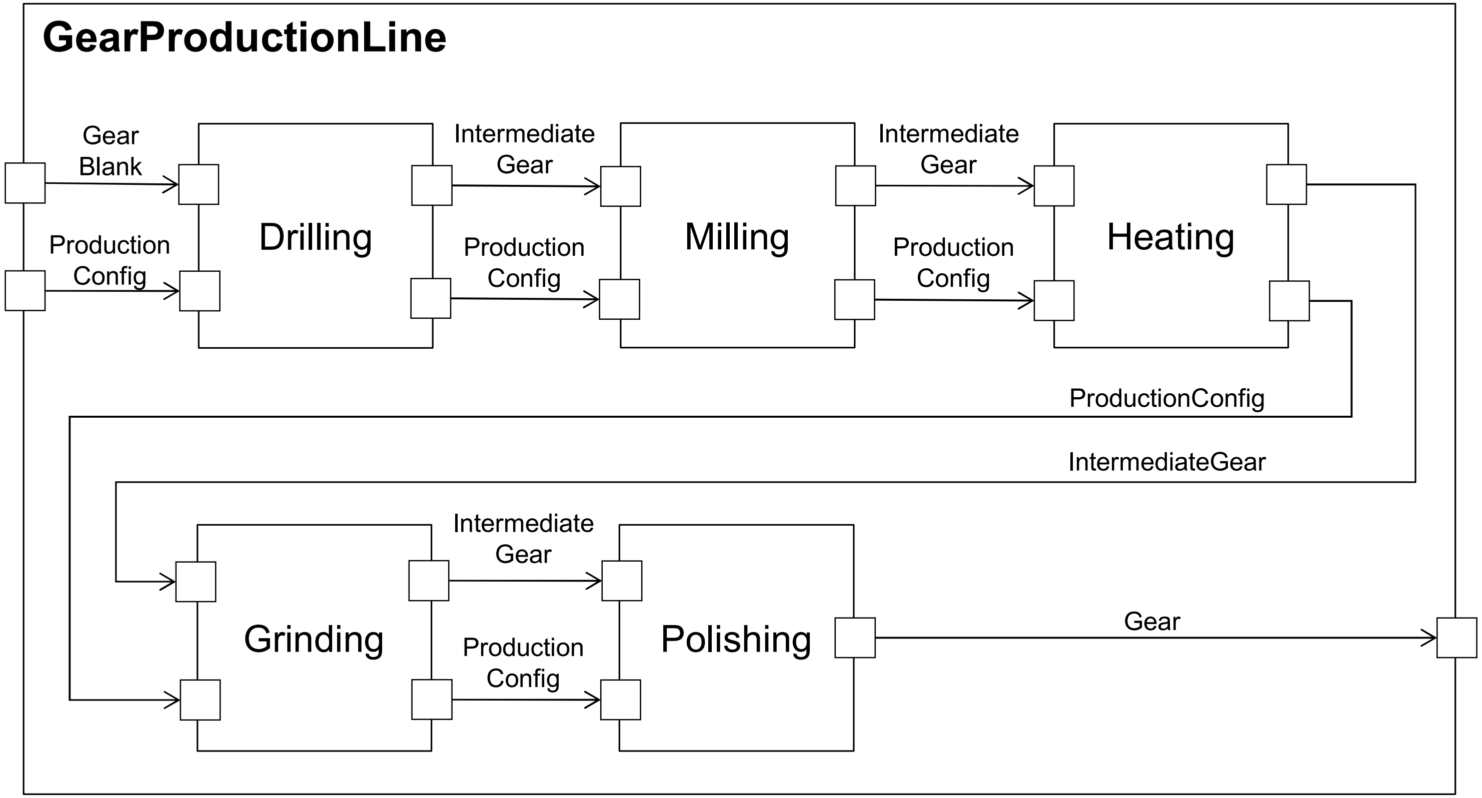


Gear Production

MA



GearProductionLine



Products

- **Gear** (https://www.gg-antriebstechnik.de/zahnraeder-stirnraeder)
  - Diameter
  - No. of Teeth
  - Material
- Wheel
- Angle / Hinge
- Other production steps:
  - Grinding
  - Polishing
- Machine configuration
  - Position
  - Duration
  - Depth (drilling)
  - Path (milling)
  - Temperature (heating)